Newton's gravitational law to ever more stringent tests. Occasionally new astronomical measurements disagreed with Newton's law, but in due course the observations or their interpretation turned out to be wrong. Time after time Newton's law triumphed over experimental or intellectual error. For example, when the motion of the planet Uranus (which had been discovered in 1781) appeared to violate the predictions of Newton's gravitational law, it seemed likely that the gravity of some other, undiscovered planet must be pulling on Uranus, perturbing its orbit. Calculations by U. J. J. Le Verrier, based solely on Newton's laws of gravity and motion and on the observations of Uranus, predicted where in the sky that new planet should be. In 1846, when J. G. Galle trained his telescope on the spot, there the predicted planet was, too dim to be seen by the naked eye but bright enough for his telescope. This new planet, which vindicated Newton's gravitational law, was given the name "Neptune."

In the early 1900s, there remained two other exquisitely small, but puzzling discrepancies with Newton's gravitational law. One, a peculiarity in the orbit of the planet Mercury, would ultimately turn out to herald a failure of Newton's law. The other, a peculiarity in the Moon's orbit, would ultimately go away; it would turn out to be a misinterpretation of the astronomers' measurements. As is so often the case with exquisitely precise measurements, it was difficult to discern which of the two discrepancies, if either, should be worried about.

Einstein correctly suspected that Mercury's peculiarity (an anomalous shift of its *perihelion;* Box 2.2) was real and the Moon's peculiarity was not. Mercury's peculiarity "smelled" real; the Moon's did not. However, this suspected disagreement of experiment with Newton's gravitational law was far less interesting and important to Einstein than his conviction that Newton's law would turn out to violate his newly formulated principle of relativity (the "metaprinciple" that all the laws of physics must be the same in every inertial reference frame). Since Einstein believed firmly in his principle of relativity, such a violation would mean that Newton's gravitational law must be flawed.[2]

Box 2.2

The Perihelion Shift of Mercury